

Research Article

Integration of IoT, Machine Learning, and Sensors for Intelligent Environmental Monitoring and Agricultural Development

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This paper presents an integrated framework for intelligent agricultural monitoring and development by combining Internet of Things (IoT) technology, machine learning algorithms, sensor networks and custom hardware design. A comprehensive system was developed using environmental sensors including DHT11, soil moisture probes, BMP180 pressure modules, MQ-4 gas detectors, rain detection sensors and HC-SR04 ultrasonic modules, interfaced via custom-designed printed circuit boards (PCBs) fabricated using Proteus software. NodeMCU ESP8266, ESP32 DevKit and ESP32-CAM microcontrollers served as the hardware backbone for real-time data acquisition, wireless transmission, and image capture. Collected sensor data were transmitted to cloud platforms through Adafruit IO for remote visualization and analysis. Machine learning models, including Random Forest and XGBoost classifiers, were trained on features extracted from VGG16-based image processing to classify plant health conditions with high accuracy. Intelligent irrigation control was achieved through autonomous decision-making based on real-time sensor feedback and environmental conditions, dynamically activating a water pump system. The integration of low-power hardware, efficient PCB layouts, cloud-based dashboards, and lightweight machine learning models resulted in a scalable, portable, and cost-effective smart farming solution. Experimental results validate the system's capability for accurate environmental monitoring, efficient resource utilization, and intelligent crop management, offering significant potential for sustainable agriculture in resource-constrained settings.

Keywords: autonomous farming; Internet of Things (IoT); machine learning (ML); real-time monitoring; sustainable agriculture

1. Introduction

With the rapid evolution of agricultural science, integrating intelligent technologies has shifted from a luxury to an operational imperative. Conventional farming, dependent on human intuition and experience, is increasingly inadequate to extract optimal yields from ever-dwindling parcels of arable land. Expanding urban footprints, driven by the need to house a growing global population, continue

to encroach on farmland; at the same time, climate patterns have become markedly more volatile and less predictable than ever before. This increasing environmental unpredictability makes agricultural decisions more complex and difficult to navigate with traditional approaches. Consequently, agriculture now demands solutions that are faster, more adaptive, and thoroughly data-driven. Although traditional farming has sustained humanity since antiquity, it no longer suffices for the multifaceted challenges of the 21st

century. In this milieu, smart agriculture—underpinned by the Internet of Things (IoT) and artificial intelligence (AI)—emerges as a forward-looking paradigm that not only satisfies immediate productivity imperatives but also aligns with intergenerational sustainability goals and environmental stewardship.

IoT-enabled agriculture capitalizes on real-time sensing to monitor field conditions continuously and unceasingly. Several research initiatives have already demonstrated the feasibility of automating agricultural processes through the deployment of IoT devices [1, 2]. Such systems track essential environmental parameters—including temperature, relative humidity, soil-moisture content [3–9], and even rainfall events [7, 8] with granular temporal resolution and high spatial fidelity. Monitoring systems based on IoT give farmers the power and freedom to make rigorously data-driven decisions [7, 9] fine-tune irrigation schedules to balance soil humidity and electrical energy consumption by controlling irrigation systems [2, 7–9] and sustain optimal microclimates for diverse crops across varied terrains. This constant stream of environmental data empowers producers to act swiftly and accurately, replacing experience-based conjecture with actionable insights and eliminating intervention delays. Precision irrigation, for example, curtails over-watering during rainy spells or periods of elevated soil humidity, thereby conserving what is rapidly becoming one of the planet's scarcest resources—water—potentially proving lifesaving for future generations. The same feedback loop mitigates nutrient runoff, directly reducing expenditures on pesticides and fertilizers and promoting ecological balance. Furthermore, remote connectivity offers ubiquitous situational awareness, allowing growers to supervise fields from anywhere through mobile applications and web interfaces. By orchestrating the judicious use of constrained resources—particularly water and electrical power—IoT solutions markedly enhance operational efficiency, a benefit that is especially salient in resource-limited or geographically isolated farming contexts.

Machine learning (ML) and deep learning (DL)—the latter a representation-learning subset of ML—construct predictive models that learn directly from data and improve over time. Recently, Raspberry Pi with ESP32 CAM has been used to analyze DL methods and IoT [10, 11]. Classical ML encompasses regression, classification, and clustering techniques, whereas DL leverages multilayered neural networks capable of extracting hierarchical representations from complex inputs, rendering it highly effective for image recognition [12–14], natural language processing [15, 16], speech analytics [17–19], autonomous navigation [20–22], and beyond. These technologies have delivered transformative outcomes across multiple sectors, including healthcare [23–25], manufacturing [26–28], and, pertinently, agriculture [29–31]. Within the agricultural domain, one of the most pressing challenges is the early detection [32–34], control [35], and mitigation [36, 37] of plant diseases—events that can devastate yields if interventions are delayed. Parallel to these imperatives, a growing body of work has applied ML and DL algorithms to diagnosing plant disease [34, 38–40] through image processing and

classification techniques, as well as to weather forecasting and irrigation optimization [9]. Conventional inspection, however, remains labor-intensive and time-consuming; it obliges farmers to scrutinize each plant manually and is therefore susceptible both to oversight and to procrastination-induced inaccuracies [41]. Disease may spread significantly by the time visual symptoms are noticed by humans, compounding losses and reducing effectiveness of intervention measures.

Integrating environmental monitoring data with disease-detection analytics offers a holistic decision-making framework. For instance, a sudden spike in humidity or soil moisture detected by IoT sensors may indicate favorable conditions for specific pathogens. Upon receiving such alerts, lightweight image classifiers employing ML/DL can rapidly analyze leaf imagery to validate whether disease is present. These classifiers can run in near-real time and are often capable of distinguishing between various types of diseases with minimal human oversight. This synergistic interplay between IoT and ML/DL enhances the system's ability to identify, predict, and respond to phytopathological threats with unprecedented speed and precision. It significantly reduces reliance on manual labor, mitigates human error, and introduces a predictive layer that traditional methods lack.

Despite significant strides in both automations via IoT and disease diagnostics via ML/DL, most existing investigations still treat these facets in isolation. Research focused solely on IoT-based monitoring lacks the diagnostic acuity offered by ML/DL, whereas ML/DL-centric frameworks frequently overlook the environmental variables that critically influence disease progression. Only a handful of studies endeavor to amalgamate real-time environmental surveillance with intelligent disease detection [42, 43], and even these typically privilege either actuation or diagnostics, rather than treating them as coequal pillars. This fragmented approach constrains the broader applicability and scalability of smart-agriculture solutions. Furthermore, lack of a closed feedback mechanism between environmental inputs and disease outputs in existing frameworks hinders rapid, coordinated responses to emerging threats.

The present study seeks to bridge that gap by introducing a unified architecture that seamlessly couples IoT-derived environmental telemetry with ML/DL-driven leaf-disease diagnostics. By fusing continuous sensor data acquisition with on-device image analysis, the proposed framework constitutes a closed-loop system that supports dynamic irrigation control, expeditious disease intervention, and evidence-based agronomic management. Through this integration, farmers receive actionable insights grounded in both ambient conditions and plant health, enabling rapid, proportionate responses to emergent threats and fostering resilient, high-yield production systems. The combination of smart sensors, cloud connectivity, edge computing, and real-time AI inference models opens the door for scalable and sustainable smart agriculture systems with broad applicability.

The remainder of this paper is organized as follows. Section 2 elaborates the system's design and

implementation, including hardware integration for environmental monitoring, the development of an automated irrigation subsystem, and the deployment of surveillance cameras for advanced ML/DL-based leaf-disease detection. Section 3 evaluates the system's performance in real-world conditions and discusses its practical applications, underscoring the framework's capacity to advance efficient and sustainable agricultural practices. Finally, Section 4 summarizes the principal findings and outlines promising directions for future research and expansion of this integrated model.

2. Materials and Methods

In our designed intelligent agricultural system, a range of sensors and actuators were integrated to enable real-time environmental monitoring and intelligent irrigation control. The DHT11 temperature and humidity sensor was used to measure ambient environmental conditions, providing digital output with acceptable accuracy for agricultural applications. Soil moisture was monitored using a capacitive soil moisture sensor, which generated analog signals proportional to the water content in the soil, facilitating automated irrigation decisions. Atmospheric pressure measurements were captured through the BMP180 barometric pressure sensor, which communicates via the I²C protocol and delivers high-resolution pressure and temperature data essential for weather prediction and environmental assessment.

A mini submersible water pump served as the irrigation actuator, automatically triggered based on soil moisture readings to maintain optimal soil hydration. To monitor gas levels and ensure safety against potential hazards, the MQ-4 gas sensor was utilized, detecting combustible gases such as methane through variations in the conductivity of a tin dioxide sensing layer. Rain detection was achieved using a rain sensor module, which provided digital signals indicating the presence or absence of surface water, preventing unnecessary irrigation during rainfall events. Furthermore, an HC-SR04 ultrasonic distance sensor was employed to measure the distance from the sensor to the soil surface, enabling real-time detection of surface conditions and irrigation effects without direct soil contact. The integration of these sensors and actuators provided a comprehensive, autonomous framework for efficient water management, environmental monitoring, and enhanced decision-making in agricultural operations. The physical modules corresponding to each sensor and actuator are shown in Figure 1.

To enhance the structural integrity, compactness, and reliability of the overall system, custom printed circuit boards (PCBs) were designed and fabricated for sensor integration and actuator control. The PCB designs were developed using Proteus Design Suite, a professional software platform widely used for schematic capture, PCB layout, and circuit simulation. Two distinct PCB boards were designed based on the system requirements as shown in Figure 2. Board 1 in Figure 2(a) integrates the DHT11 temperature and humidity sensor, DS18B20 soil temperature sensor, and soil moisture sensor, interfacing them with

a NodeMCU V3 microcontroller. This board is also responsible for controlling the water pump through a TIP122 transistor switch configuration. The power regulation for the system was managed by a 7805 voltage regulator, ensuring stable 5 V output for sensor operation.

Board 2 in Figure 2(b) was designed to accommodate the BMP180 pressure sensor, gas sensor, rain detection module, and HC-SR04 ultrasonic sensor which is repurposed for water level detection. This board similarly interfaces all sensors with a second NodeMCU V3 microcontroller and includes additional components such as a buzzer and LED indicators to signal abnormal environmental conditions such as high gas concentration or water detection events. Both PCBs feature regulated power supplies, input/output pin headers, and appropriate passive components such as resistors and capacitors to stabilize sensor readings and prevent electrical noise. After successful simulation and verification of the circuit designs in Proteus, the PCB layouts were finalized and printed. The detailed PCB layouts and 3D rendering diagrams have been provided in Figures S1 and S2 in the Supporting Information. The adoption of PCBs significantly improved the durability and organization of the system, reduced wiring complexity, and minimized the risk of loose connections and interference, leading to a more robust and reliable smart agriculture platform. Furthermore, multiple microcontroller platforms were utilized to achieve diverse functionalities including sensor interfacing, wireless communication, data collection, and image capture. The ESP32 DevKit V1 board in Figure 3(a) served as the primary microcontroller for managing complex tasks requiring higher computational performance and multiple simultaneous sensor inputs. Featuring a dual-core processor, integrated Wi-Fi and Bluetooth capabilities, and ample GPIO availability, the ESP32 DevKit V1 provided robust support for real-time environmental monitoring, data processing, and cloud communication tasks.

The NodeMCU ESP8266 board in Figure 3(b) was employed for lightweight sensing and control operations where moderate computational resources were sufficient. Due to its integrated Wi-Fi module, low power consumption, and cost-effectiveness, the NodeMCU was ideal for continuous sensing of environmental parameters such as temperature, humidity, soil moisture, and transmitting the collected data to cloud platforms such as Adafruit IO. Additionally, the ESP32-CAM module in Figure 3(c) was incorporated for image-based monitoring applications. This board integrates an ESP32 SoC with a camera module and onboard Wi-Fi connectivity, allowing real-time image capture, edge processing, and wireless transmission without the need for an external processor. Its compact form factor and camera integration made it highly suitable for remote plant health observation and security monitoring within the smart agricultural system. Together, the deployment of these microcontroller boards enabled a scalable, flexible, and highly connected system architecture capable of supporting a variety of smart farming applications, ranging from environmental sensing to visual inspection. The complete hardware setup of the proposed smart agricultural system is illustrated in Figure 4. The front view of the system is shown

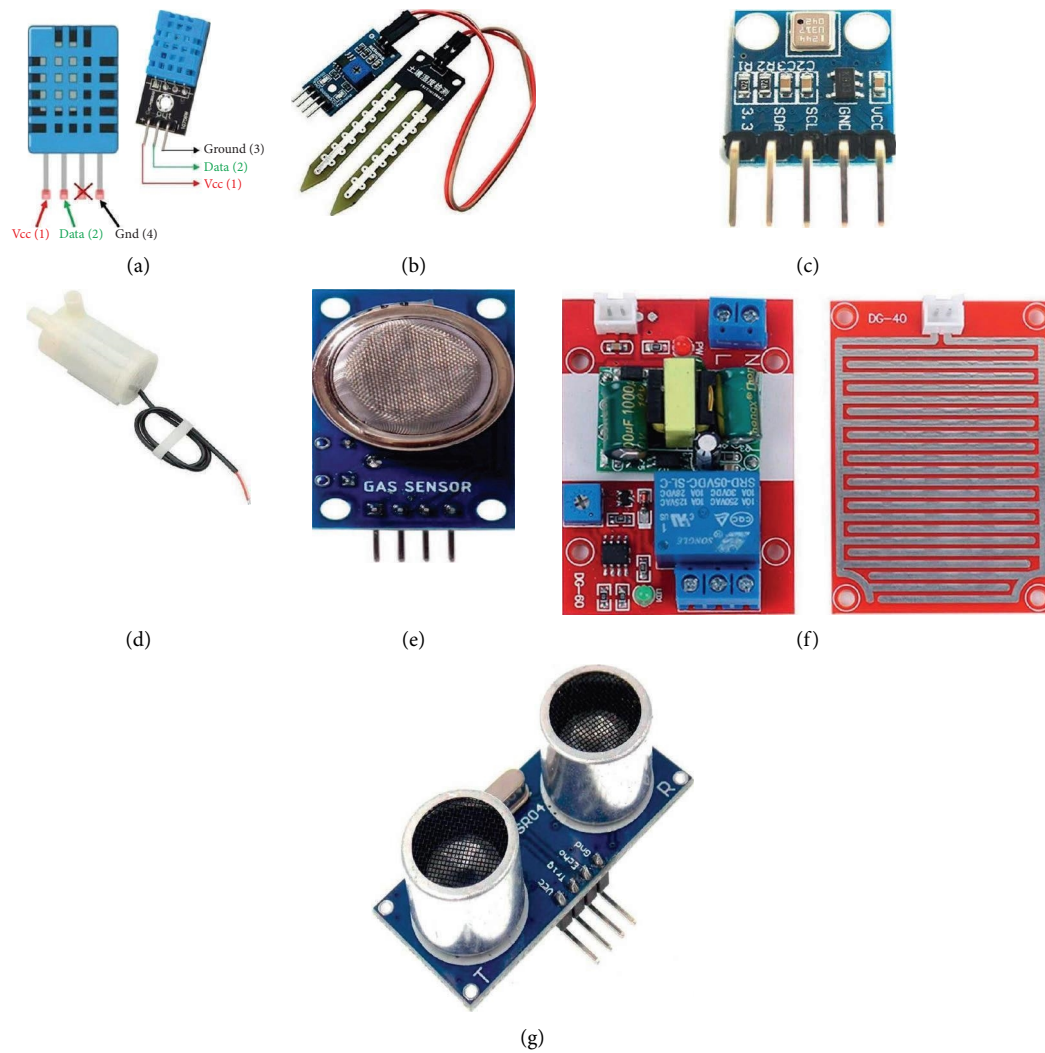
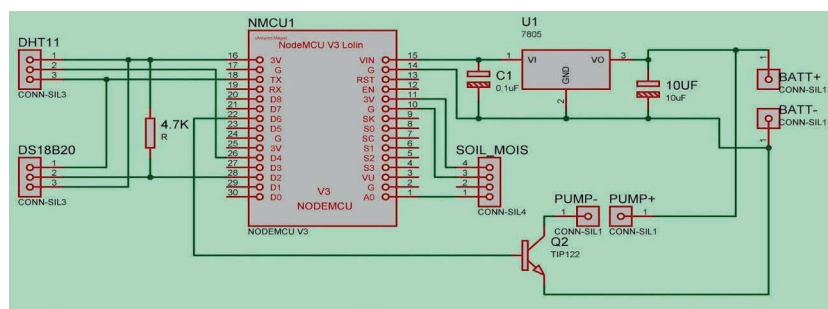


FIGURE 1: Sensors and modules used in the system. (a) DHT11 sensor, (b) soil moisture sensor, (c) pressure sensor, (d) mini water pump, (e) gas sensor, (f) rain sensor module, and (g) ultrasonic sensor.



(a)

FIGURE 2: Continued.

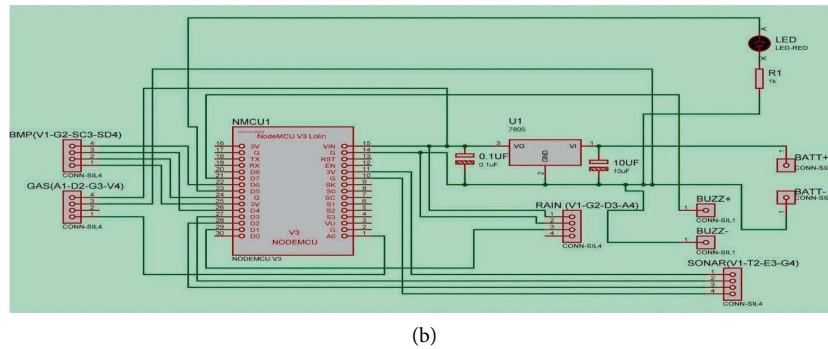


FIGURE 2: Circuit diagrams of the developed system: (a) Board 1 integrating DHT11, DS18B20, soil moisture sensor, and water pump control; (b) Board 2 integrating BMP180, gas sensor, rain sensor, ultrasonic sensor, and buzzer module.

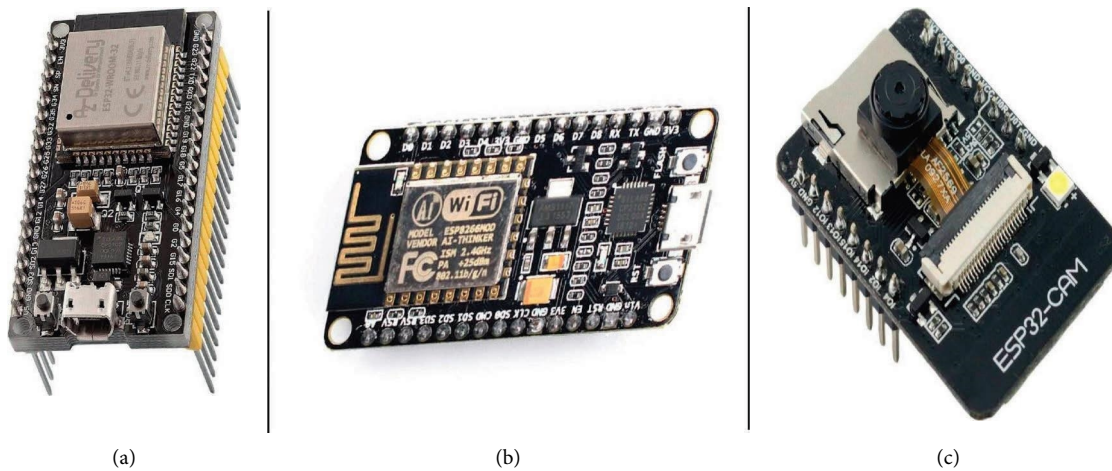


FIGURE 3: Microcontroller boards used in the system: (a) ESP32 DevKit V1, (b) NodeMCU ESP8266, and (c) ESP32-CAM module for wireless sensing and image capturing applications.

in Figure 4(a), while Figure 4(b) presents a top-down perspective. The experimental platform consists of live plants embedded within a soil-filled transparent container, over which multiple sensors and electronic modules are deployed to monitor environmental conditions and automate irrigation responses. Custom designed PCBs, PCB Board 1, and PCB Board 2 were fabricated and mounted adjacent to the planting area. PCB Board 1 interfaces with the soil moisture sensor and the DHT11 temperature-humidity sensor, providing real-time environmental monitoring. It also controls the activation of the water pump based on soil moisture levels. PCB Board 2 integrates additional modules including the MQ-4 gas sensor for air quality monitoring and a rain detection sensor for rainfall identification, contributing to more intelligent irrigation decision-making.

An HC-SR04 ultrasonic distance sensor was positioned near the soil surface to monitor soil water levels non-invasively by measuring the distance between the sensor and the soil. The ESP32-CAM module, as depicted in Figure 4(b), was mounted vertically on a wooden structure above the plants to capture periodic images for visual

health monitoring and remote observation via Wi-Fi connectivity. A portable power bank was used to provide an independent and continuous power supply to the entire system, enhancing its deployability in off-grid agricultural environments. Careful cable management and modular placement ensured system stability, minimal interference among components, and ease of maintenance. This integrated arrangement of sensors, microcontrollers, actuators, and cloud connectivity components forms a comprehensive smart farming solution capable of real-time monitoring, autonomous irrigation, environmental safety assurance, and remote visual inspection. Complete PCB packaging has been provided in Figures S3 and S4, and detailed cost analysis has been provided in Table S1 in the Supporting Information. Figure 5 illustrates the overview of the architecture of the designed system and highlights the coordination between various integrated units.

3. Results and Discussion

In this section, we detail the results achieved utilizing the approaches outlined above, as well as their implementation

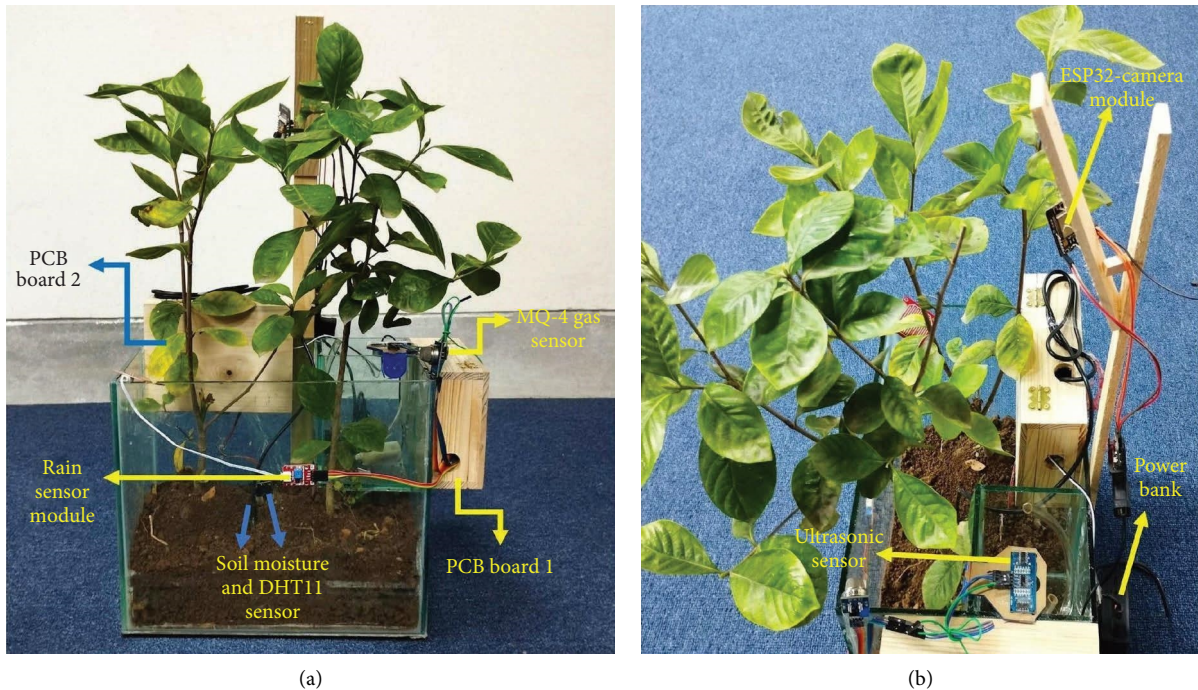


FIGURE 4: Complete experimental setup of the intelligent agriculture system showing (a) front view with sensor modules and PCBs and (b) top view highlighting the ESP32-CAM module, ultrasonic sensor, and power supply arrangement.

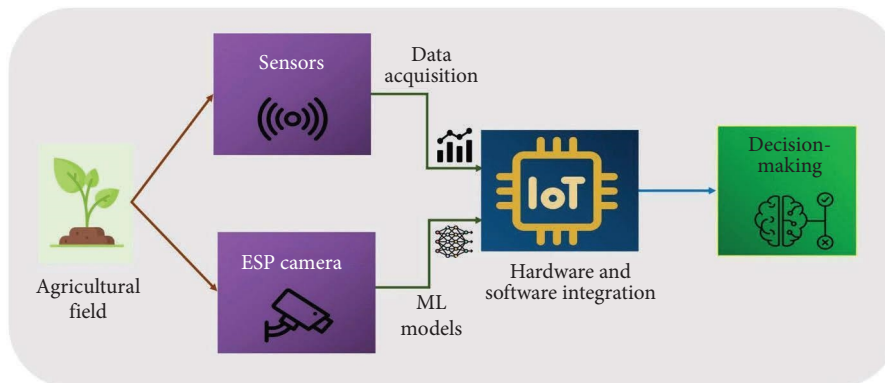


FIGURE 5: Comprehensive workflow for smart agricultural monitoring and development.

and potential applications. The performance of the integrated sensor network, the effectiveness of the PCB hardware design, and the accuracy of the ML models are evaluated and discussed. Real-time monitoring capabilities, system responsiveness under different environmental conditions, and resource optimization outcomes are presented.

3.1. Cloud Data Monitoring. To enable centralized monitoring and control within our intelligent farming framework, we developed a comprehensive cloud-based dashboard using Adafruit IO as shown in Figure 6, which visualizes both real-time sensor data and historical trends [44]. The ESP32 microcontroller continuously collects environmental and plant-soil parameters, including ambient temperature, soil temperature, humidity, soil moisture, and gas concentration, and transmits them via MQTT protocol

[45] to the Adafruit IO server. These data streams are displayed in a clean, interactive format that allows users to remotely access and assess field conditions at a glance. In addition to passive monitoring, the system incorporates remote actuation capabilities, enabling users to turn on/off the water pump and LED indicators directly from the dashboard. This level of control is particularly valuable in managing irrigation based on real-time moisture levels, reducing water waste and optimizing crop health. The dashboard also includes 30-day historical plots of soil temperature and moisture, providing a time-series view that aids in identifying long-term trends or anomalies. An integrated weather forecast module further enhances decision-making by allowing users to correlate sensor data with upcoming climate conditions. The visual interface is accessible through any internet-enabled device,

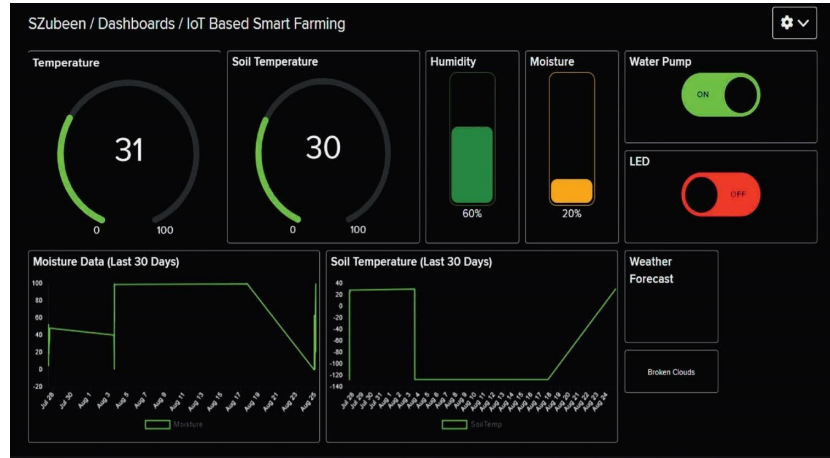


FIGURE 6: IoT-enabled smart farming interface using Adafruit IO for real-time sensing, control, and data visualization.

offering flexibility and ease of use for farmers, researchers, or agricultural technicians. This integration of edge sensing, cloud communication, and real-time visualization demonstrates the practical viability of IoT technologies in enabling data-driven precision agriculture while reducing reliance on manual labor and local infrastructure.

3.2. Real-Time Environmental Monitoring and Automated Irrigation. To facilitate real-time data acquisition and remote monitoring, Adafruit IO was utilized. Sensor data collected by the DHT11 module, including temperature and humidity measurements, were transmitted via Wi-Fi to the Adafruit IO server [46]. The platform enabled efficient storage, visualization, and retrieval of environmental data through customizable dashboards and real-time feeds. Adafruit IO's seamless integration with hardware components and its support for MQTT and REST APIs provided robust communication between the sensor node and the cloud interface. This cloud connectivity allowed for continuous monitoring and analysis of field conditions, forming the basis for intelligent irrigation decisions and system optimization.

To detect the presence of water in the soil or on surfaces, a rain sensor module was integrated into the proposed intelligent irrigation system [47]. The rain sensor consists of a PCB with parallel conductive traces designed to sense water droplets through changes in electrical conductivity [48]. In the absence of water (dry conditions), the resistance between the traces remains high, resulting in a low or inactive output signal. Conversely, when water is present, the conductivity between the traces increases, leading to a high signal output.

The module used comprises two primary components: a sensing plate and a control board. The control board features an adjustable potentiometer for sensitivity tuning and provides both analog (AOUT) and digital (DOUT) outputs, enabling flexible interfacing with microcontrollers. In this study, the digital output was monitored, where the activation of onboard LEDs indicated water status: a red LED for "No Water Detected" and a green LED for "Water

Detected," as illustrated in Figures 7(a)–7(f). During dry soil conditions, the rain sensor output remained low, which was utilized to automatically activate the water pump. Upon detecting moisture due to rain or irrigation, the output transitioned to a high state, signaling the system to deactivate the pump, thus preventing overwatering. The rain sensor module is characterized by its low cost, ease of integration, and suitability for open-field agricultural applications where precise soil moisture measurements are not critical. However, for applications requiring higher accuracy and longer term deployment, the use of capacitive soil moisture sensors could be considered as a future enhancement.

To enable real-time monitoring of flammable gas presence in the environment, an MQ-4 gas sensor was integrated into the proposed system. The MQ-4 is a semiconductor-type gas sensor specifically designed to detect methane (CH_4), natural gas, and other combustible gases [49].

It operates based on the change in the resistance of a tin dioxide (SnO_2) sensing layer in the presence of target gases. Under clean air conditions, the sensor's resistance is high, while exposure to combustible gases decreases the resistance, generating a corresponding analog signal output. In the implemented setup, the MQ-4 sensor provided continuous analog readings of gas concentrations, which were processed and transmitted to the Adafruit IO platform for real-time visualization. Figure 8 (top and bottom) demonstrates the sensor's behavior: under normal conditions, the gas concentration remained low (approximately 29%), whereas the deliberate introduction of a flame near the sensor rapidly increased the detected gas percentage to 56%. This substantial change validates the sensor's sensitivity and responsiveness to combustible gas presence. The MQ-4 sensor was selected for its low cost, high sensitivity to methane, fast response time, and ease of interfacing with microcontrollers.

The BMP180 digital pressure sensor was incorporated into the system to measure atmospheric pressure in real time as shown in Figure 9. The sensor operates via an I²C interface and provides high-resolution pressure data, which is transmitted to the Adafruit IO cloud for monitoring. Its

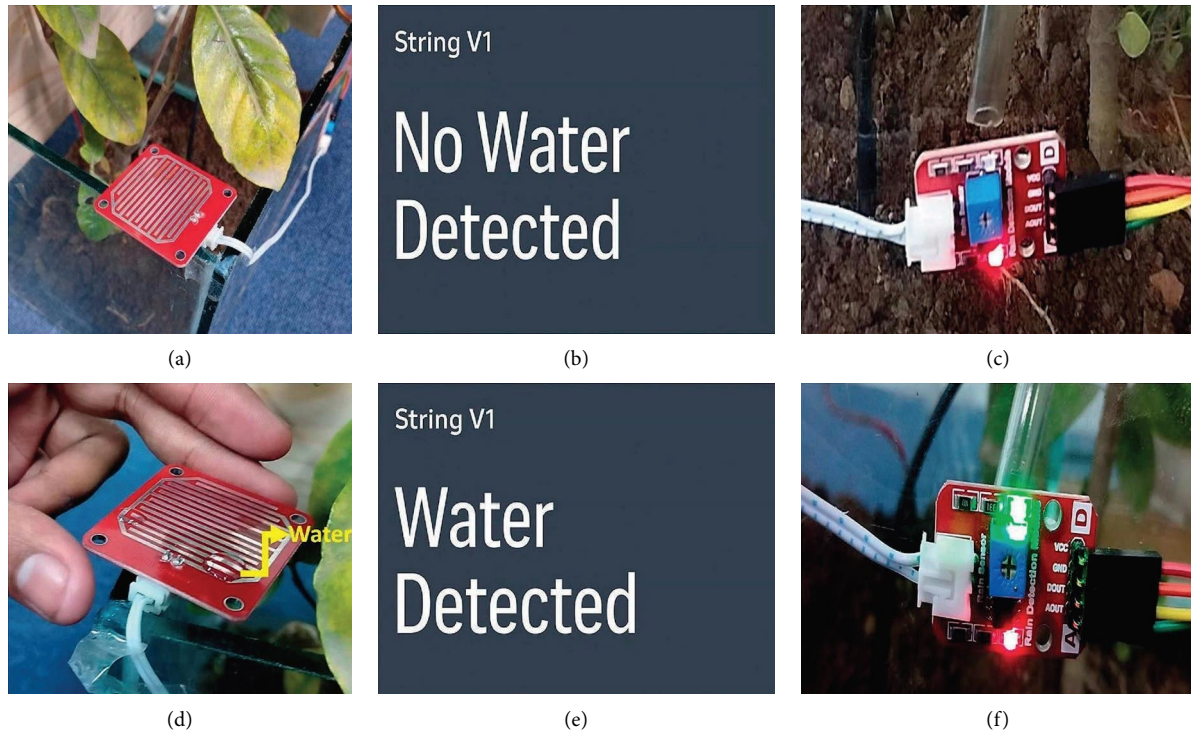


FIGURE 7: Detection results of the rain sensor module showing (a–c) dry condition with no water detected and red LED indication and (d–f) wet condition with water detected and green LED indication.

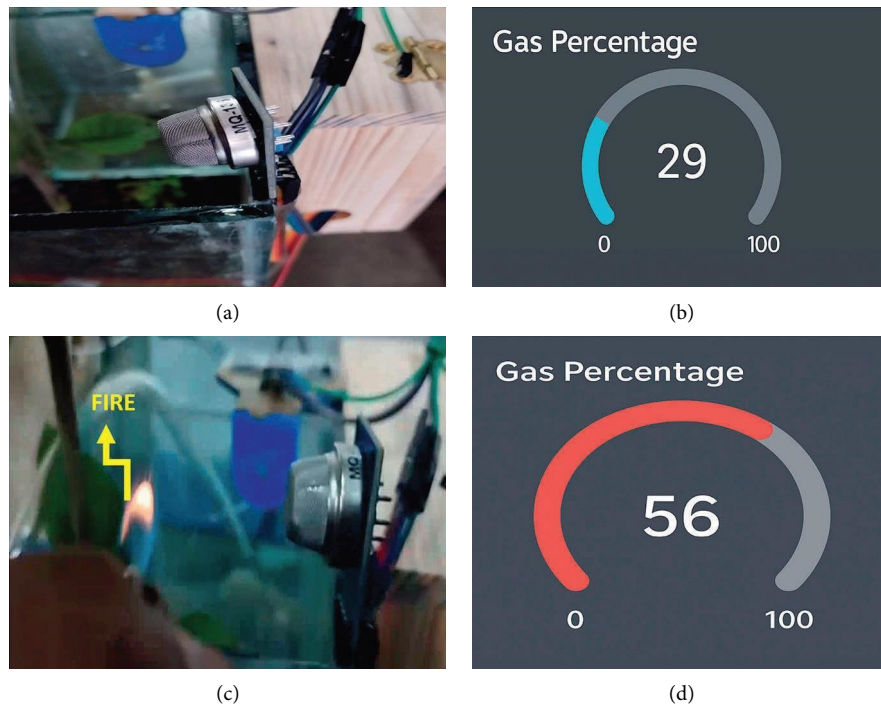


FIGURE 8: Response of the MQ-4 gas sensor showing (top) low gas concentration under normal conditions and (bottom) increased gas concentration when exposed to a fire source.

small size, low power consumption, and precision make it well-suited for smart agriculture applications where environmental sensing plays a critical role.

To monitor soil surface conditions and water levels, an HC-SR04 ultrasonic distance sensor was employed in the proposed system [50]. The HC-SR04 operates by emitting an

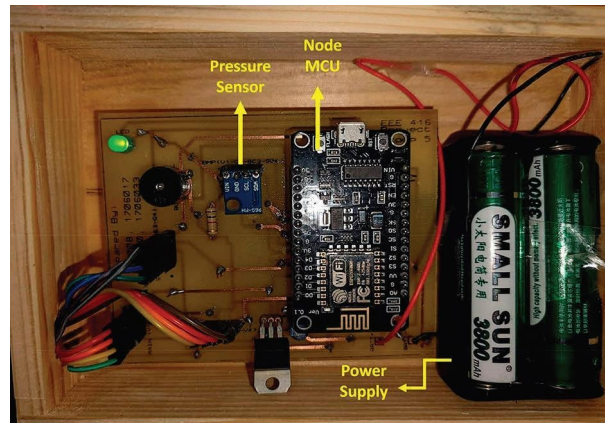


FIGURE 9: Fabricated PCB board integrating the BMP180 pressure sensor, NodeMCU microcontroller, and battery power supply for environmental monitoring applications.

ultrasonic pulse and measuring the time interval between the emission and reception of the reflected signal from a surface. Based on the measured time delay, the distance to the surface is calculated with high precision. As illustrated in Figure 10(a), the ultrasonic sensor was mounted above the soil surface to measure the vertical distance to the ground. The arrow in Figure 10(b) highlights the distance being measured between the sensor and the soil surface. Under normal dry conditions, the measured distance remained constant, corresponding to the height between the sensor and the soil bed.

When irrigation was required, triggered either by sensor readings, such as moisture sensors or threshold monitoring, the system activated the water pump. Figure 10(c) shows the sensor positioned closer to the soil where variations in soil surface or pooling water could influence the detected distance. Upon activation of the water pump, as seen in Figure 10(d), water was delivered into the soil, visibly altering surface conditions and reducing the dry distance value over time. The HC-SR04 ultrasonic sensor was selected due to its low cost, ease of interfacing, high sensitivity to distance changes, and robust performance in noncontact measurement environments. This noninvasive sensing approach supports real-time monitoring of irrigation effects without the need for direct soil contact, contributing to the development of an efficient and intelligent irrigation management system.

3.3. Leaf Image Analysis Using ML. This study involves the implementation of a Random Forest classifier [51] and an XGBoost classifier [52], utilizing features obtained from the VGG16 model [53] for training purposes. Random Forest builds many decision trees and outputs the mean prediction. It is also robust against overfitting and can handle high-dimensional data. XGBoost is another ensemble method which builds trees sequentially and is more optimized than other boosting methods. We compared these two classifiers and chose the better accuracy one for our applications. The

VGG16 model is a convolutional neural network (CNN) architecture which involves 16 layers [53]. It is mainly used to extract features from leaf images.

The performance comparison between XGBoost and Random Forest classifiers trained on VGG16-extracted features clearly demonstrates the superiority of XGBoost across all evaluated metrics as illustrated in Figure 1. XGBoost achieved an overall accuracy of 93%, significantly outperforming Random Forest, which attained only 76%. In particular, XGBoost consistently delivered high precision (≥ 0.85), recall (≥ 0.75), and $F1$ -scores (≥ 0.80) across all classes, including challenging disease categories such as Early Blight and Septoria Leaf Spot. Random Forest, in contrast, showed poor classification performance for Early Blight (precision of 0.47 and recall of 0.39) and moderate results for other disease types. The weighted average $F1$ -score for XGBoost was 0.93, compared to only 0.76 for Random Forest. These results confirm that integrating deep feature extraction via VGG16 with the XGBoost classifier provides a robust and accurate framework for the detection and classification of tomato plant diseases, outperforming traditional ensemble methods like Random Forest. The model performance is illustrated in Figure 11.

Figure 12 illustrates sample classification outputs of tomato leaf images using the proposed VGG16-based feature extraction model. Healthy leaves, characterized by consistent green coloration and intact structure, were accurately identified across multiple samples. Similarly, disease-specific symptoms such as the circular lesions of Septoria leaf spot and the yellowing indicative of spider mite infestation were correctly classified, demonstrating the robustness of the model for diverse disease manifestations. Early blight images, showing concentric dark lesions, were classified with moderate confidence, aligning with the observed dip in precision and recall for this class. Late blight samples, marked by dark water-soaked lesions, were successfully recognized, further validating the model's efficacy. Overall, the results support the model's strong performance in distinguishing between healthy and diseased leaves,

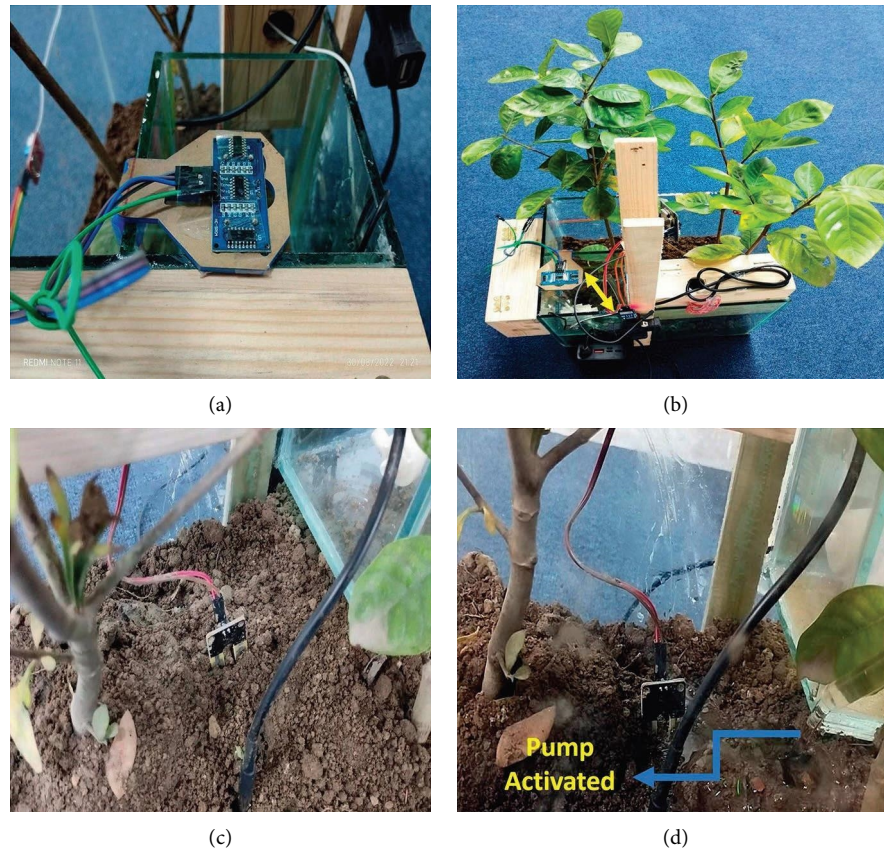


FIGURE 10: Operation of the ultrasonic sensor module showing (a) sensor placement above soil, (b) measurement setup over the plants, (c) dry soil condition before irrigation, and (d) soil condition after pump activation and water delivery.

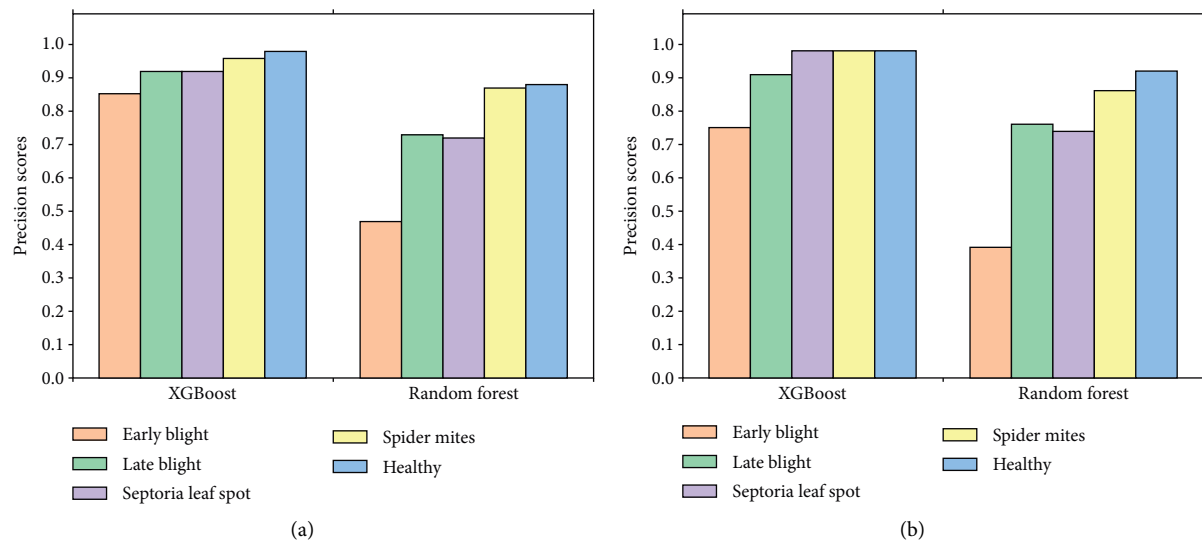


FIGURE 11: Continued.

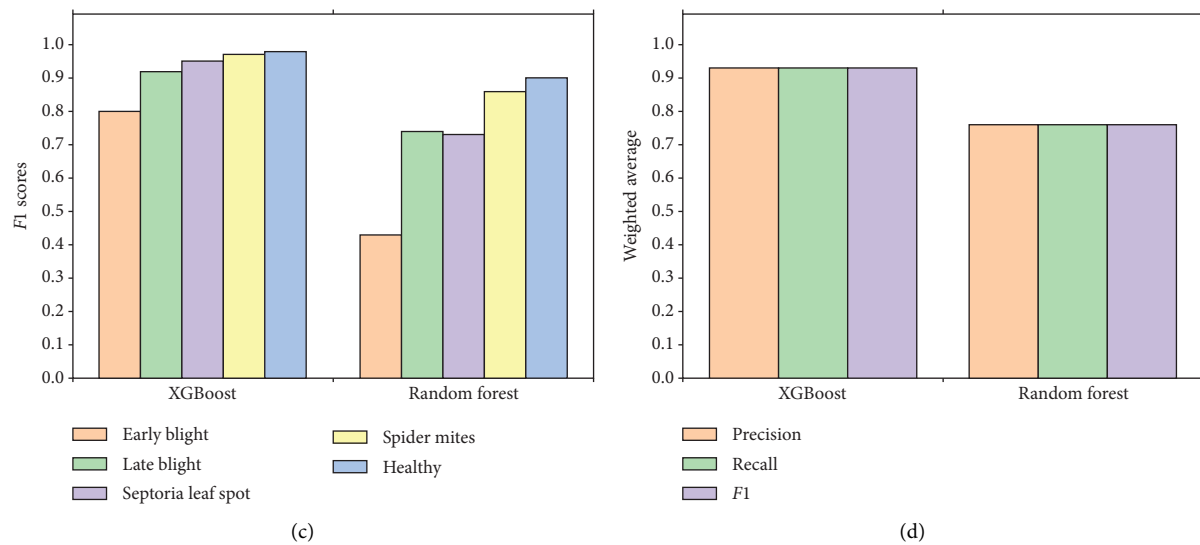


FIGURE 11: Model performance metrics comparison on tomato disease classification.

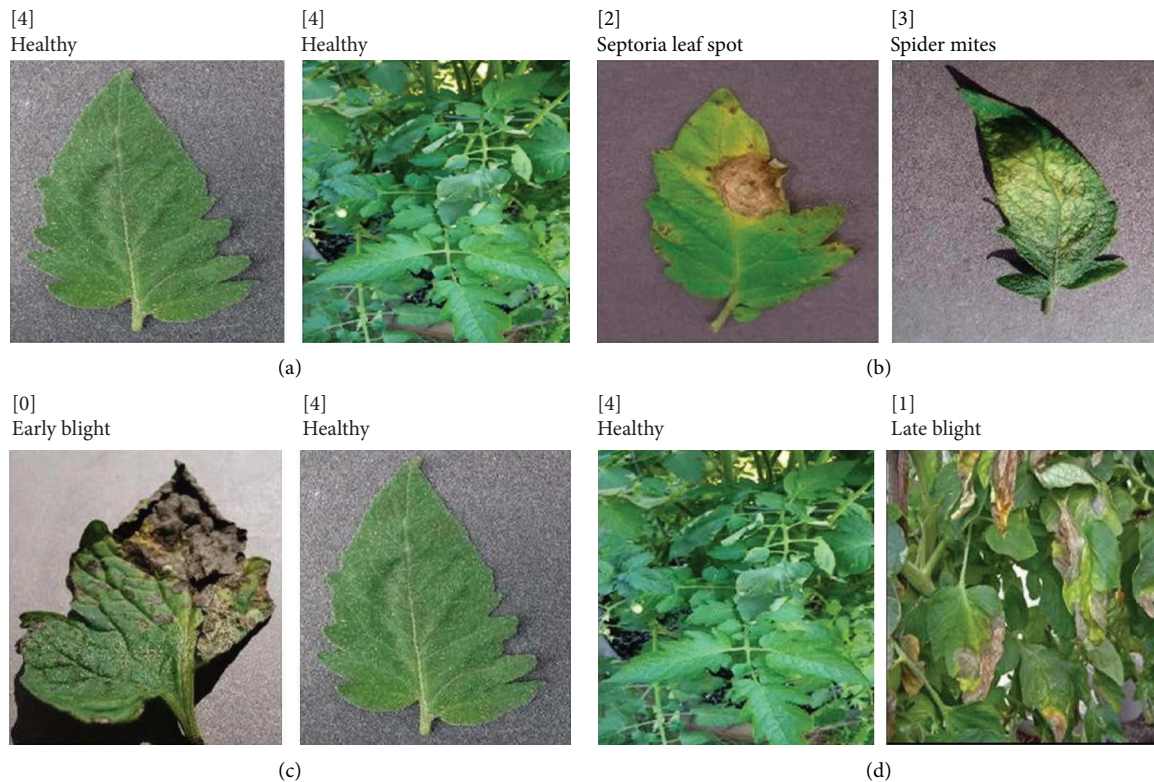


FIGURE 12: Accurate prediction of diseased and healthy leaves using the model.

although challenges remain for diseases with visually similar symptoms. Further, Figure 13 demonstrates the results of our leaf disease detection framework, which integrates an ESP32-CAM module for real-time image capture with a hybrid VGG16 + XGBoost ML pipeline. In this system, the ESP32-CAM wirelessly transmits captured images of tomato leaves to a central processing unit where deep features are

extracted using a pretrained VGG16 CNN. These features are then classified using an XGBoost model trained on a labeled dataset of tomato plant diseases. The figure showcases three classification outputs: (i) healthy leaf, (ii) Tomato Septoria leaf spot, and (iii) Tomato Spider mites. Each predicted label corresponds closely with the visible symptoms in the images, demonstrating the system's

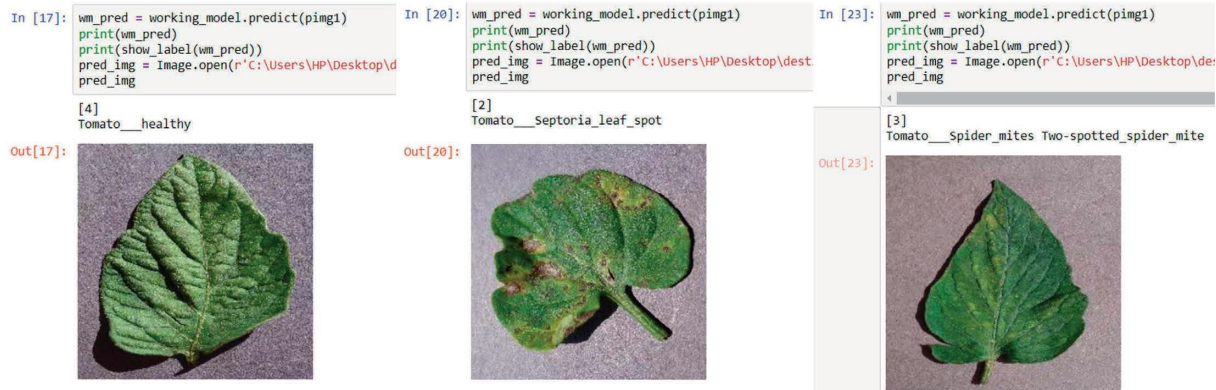


FIGURE 13: Classification results of tomato leaf health using ESP32-CAM image acquisition and VGG16-XGBoost model.

TABLE 1: Distribution of training and testing images for the selected five classes of tomato leaves from the PlantVillage dataset.

Class	Class no	Train	Test
Early blight	0	800	200
Late blight	1	1527	382
Septoria leaf spot	2	1417	354
Two spotted spider mite	3	1341	336
Healthy	4	1273	318
Total	—	6358	1590

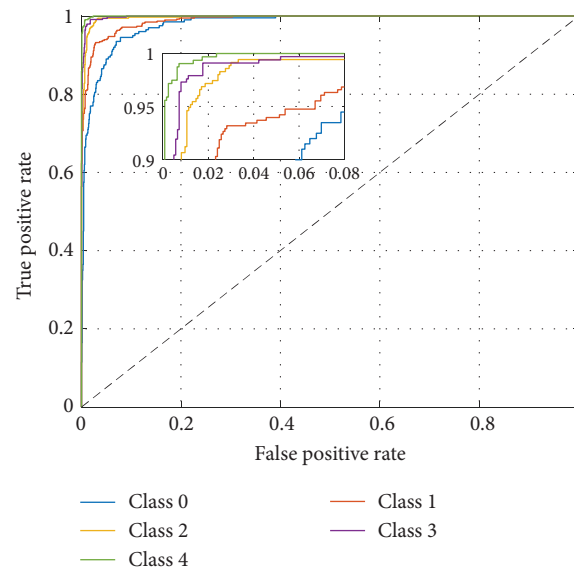


FIGURE 14: Multiclass ROC curve with corresponding AUC score for the model on the test dataset.

capability to distinguish between healthy and diseased leaves with high confidence. This approach leverages the lightweight and cost-effective nature of IoT hardware while utilizing the robustness of DL and ensemble methods for accurate, scalable, and efficient plant disease monitoring in agricultural environments.

We utilized two publicly available datasets for training and evaluating the ML models: the PlantVillage dataset and the Cassava Leaf Disease Classification dataset. The PlantVillage dataset contains 54,035 images across 14 plant species and 38 classes. Due to hardware resource constraints, we selected five classes of tomato leaves (Early Blight, Late

Blight, Septoria Leaf Spot, Two-Spotted Spider Mite, and Healthy). The dataset was split into 6358 training images and 1590 test images. The five classes in the dataset, their assigned numbers in the model, and the number of images in the training and test sets are given in Table 1:

Since the visual traits of healthy leaf images often resemble those of different species, the Cassava Leaf Image dataset was integrated into the training process to enhance the model's ability to detect healthy leaves across multiple species. However, a notable limitation remains that there is currently no publicly available dataset containing images of multiple leaves exhibiting disease symptoms, which

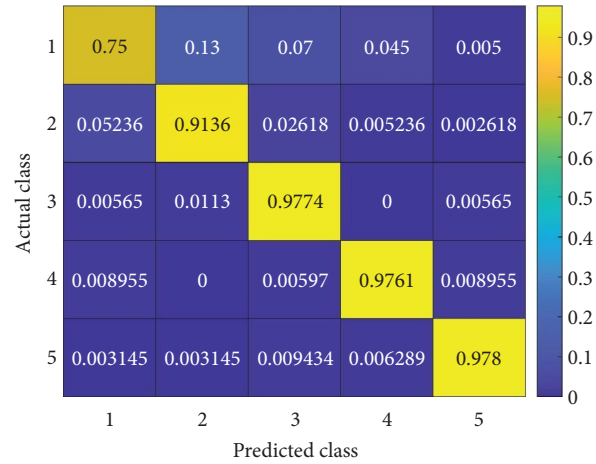


FIGURE 15: Normalized confusion matrix of the proposed deep learning model on the test dataset.

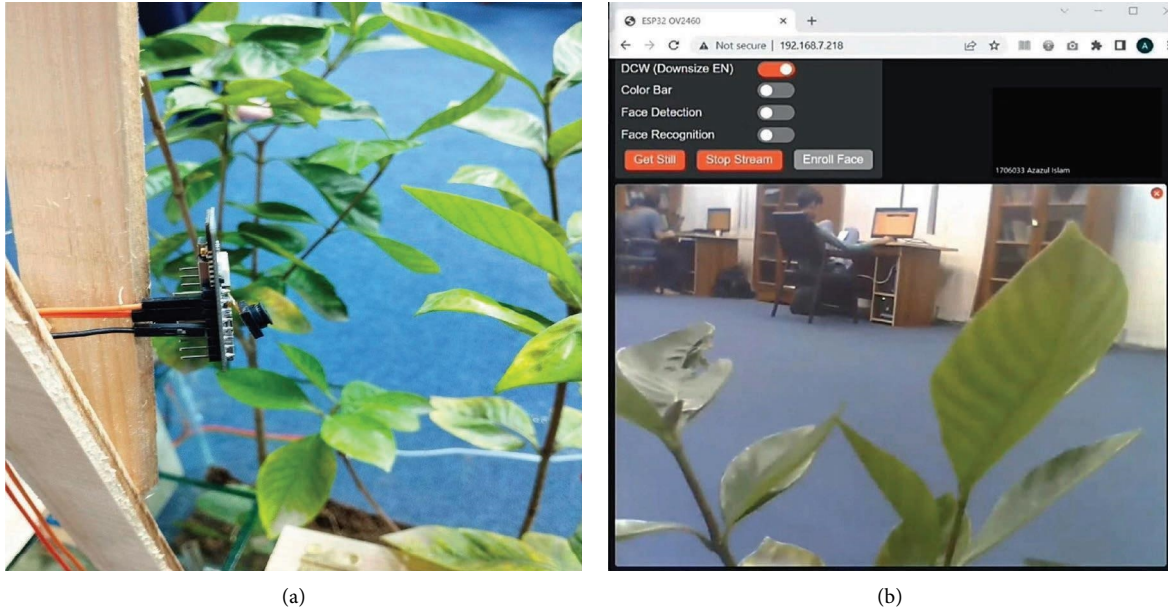


FIGURE 16: (a) Placement of ESP32-CAM for surveillance and image processing, and (b) real-time video footage view from ESP32-CAM module.

constrains the model's robustness in such scenarios. The detailed descriptions and access links for both datasets are provided in the Data Availability Statement at the end of this paper. All images from the PlantVillage and Cassava datasets were preprocessed prior to model training. Each image was resized to 224×224 pixels to match the input dimensions required by the VGG16 feature extractor. Pixel values were normalized to the range $[0, 1]$ for computational efficiency and to accelerate convergence during training. The combined dataset was divided into 70% training, 15% validation, and 15% testing. A stratified sampling strategy was employed to ensure balanced representation of all classes across the splits. For consistency, the data were organized into class-specific directories compatible with the Keras/TensorFlow data pipeline. The VGG16 CNN pre-trained on ImageNet was employed as a fixed feature

extractor. The network consists of five convolutional blocks, each comprising multiple convolutional layers with 3×3 filters and ReLU activation, followed by max-pooling layers for spatial downsampling. The final block outputs feature maps of size $8 \times 8 \times 512$, capturing high-level image representations. In total, the model contains approximately 14.7 million parameters, all of which were kept nontrainable to preserve the prelearned weights and reduce computational cost. These extracted features were subsequently fed into ensemble classifiers (Random Forest and XGBoost) for plant disease classification, combining the representational strength of DL with the robustness of traditional ML algorithms.

Figure 14 shows the receiver operating characteristic (ROC) curves for the five-class tomato classification problem. Each curve corresponds to one of the classes (Class 0–4), plotted as the true positive rate against the false

TABLE 2: Comparison of ML/DL techniques and features in agricultural automation.

Serial	ML/DL technique (model used)	Plant	Features	Accuracy	Reference
1	—	—	WiFi integration, IoT-based automation	No ML part	[55]
2	—	—	WiFi/GSM integration, IoT-based automation, SMS notification, weather forecast	No ML part	[56]
3	—	—	Same as above	No ML part	[57]
4	Decision tree algorithm	No specific plant	Used ML decision tree algorithm to take decisions to predict moisture, temperature, humidity, and the water necessity.	No accuracy value reported	[58]
5	SVM classifier	Rice	Machine vision, SVM classification, GA-based feature selection, rice seedling Bakanae disease detection.	87.9%	[59]
6	Machine vision system using hyperspectral imaging (HSI)	Chili pepper	Hyperspectral imaging, UV + halogen illumination, feature selection (FDP, PCA, SVM-RFE, HBBE), MLP/LDA classification for aflatoxin detection in chili peppers.	87.5%	[60]
7	K-means clustering algorithm, SVM	Papaya	Mobile-based image acquisition, k-means segmentation, statistical + GLCM feature extraction, SVM classification for papaya disease recognition.	90.15%	[61]
8	K-means algorithm, multi SVM	Soyabean culture	K-means segmentation, color + texture feature extraction, rule-based classification, multi-SVM models for soybean leaf disease detection and severity estimation.	90%	[62]
9	K-means clustering, SVM	Grape leaf	Image preprocessing (resizing, enhancement, smoothing), LAB + HSI color transformation, K-means segmentation, GLCM + statistical feature extraction, multi-class SVM classification	82.5	[63]
10	VGG-16	No specified plant	Image preprocessing, ML-DL based disease detection using RF, SGD, SVM, VGG-19, Inception-v3, VGG-16	89.5	[64]
11	Deep convolutional neural network (DCNN)	Winter wheat	UAV-based hyperspectral imaging, deep CNN with Inception-ResNet layers, spatial-spectral feature fusion, ML/DL-based yellow rust detection in wheat	85	[65]
12	Image processing, K means clustering, SVM	Grapes	Grape leaf disease detection of illnesses that affect leaves include powdery and downy mildew using SVM classification, K-means segmentation, and combined color & texture feature extraction	88.89	[66]
13	Fuzzy rule-based approach for disease detection (FRADD)	Apples	Illness of fruits detection using fuzzy rule-based classification with blob analysis for feature extraction	91.66	[67]
14	SVM with quadratic kernel	Precocious dwarf cashew and coconut	Hydra multilevel data fusion, outlier detection & signal filtering, decision fusion across applications, smart agriculture evapotranspiration modeling with ML	RMSE = 0.79 $R^2 = 0.74$ Correlation with Penman-Monteith reference model = 0.9299	[68]

TABLE 2: Continued.

Serial	ML/DL technique (model used)	Plant	Features	Accuracy	Reference
15	CNN with K means clustering	Weed	Soybean weed identification, K-means unsupervised feature learning, CNN-based deep learning, optimized weight initialization	92.89%	[69]
16	CNN	Grape	PlantVillage dataset, grape leaf disease detection, custom real-life image database, deep siamese CNN	90%	[70]
17	Random forest	Maize leaf	Maize leaf disease detection, image preprocessing and feature extraction, ML classifiers (NB, KNN, DT, SVM, RF), Random Forest	79.23%	[71]
18	AlexNet with kNN classifier	Tomato	Tomato leaf disease detection and classification using CNN-based feature extraction (AlexNet) and kNN classifier	76.1%	[72]
19	Deep CNN, modified Inception-ResNet	Tomato	Deep learning with synthetic image training, real-time fruit counting, robust to occlusion, scale, and illumination variation.	91%	[73]
20	Deep learning	Strawberry	UAV-based orthoimage acquisition, faster R-CNN with ResNet-50 for strawberry flower and fruit detection, and yield prediction mapping.	84.1%	[74]
21	CNN & LSTM	Winter wheat	Remote sensing integration, meteorological and soil data fusion, yield detrending, deep learning-based wheat yield prediction using LSTM + CNN with uncertainty quantification	84%-85%	[75]
22	Random Forest, XGBoost classifiers, VGG-16	Tomato	WiFi integration, IoT-based automation, ML DL based disease detection using XGBoost	93%	This work

positive rate. All curves are well above the diagonal line, indicating that the model performs substantially better than random guessing. The inset provides zoomed-in view of the high-performance region (false positive rate < 0.1), where the curves remain close to the upper-left corner, confirming excellent discriminative ability across all classes. The area under the curve (AUC) values for all classes exceed 0.9, reflecting strong predictive power. Figure 15 presents the normalized confusion matrix for the five-class classification problem.

The diagonal entries correspond to the proportion of correctly classified samples for each class, while the off-diagonal values represent misclassifications. The model achieves very high accuracy for Classes 3, 4, and 5, with values of 0.977, 0.976, and 0.978, respectively. Class 2 is also classified with strong accuracy (0.914). However, performance for Class 1 is comparatively lower (0.75), with notable misclassifications into Class 2 (0.13) and Class 3 (0.07). This indicates that the model has some difficulty distinguishing Class 1 from its neighboring categories, while other classes remain well separated.

3.4. Security Measurements. In addition to its role in plant health monitoring, the ESP32-CAM module was also configured as a low-cost, wireless security surveillance system within the smart farming framework as shown in Figure 16. Positioned strategically within the agricultural field, the module continuously captures real-time video or periodic still images, which are transmitted over to a secure cloud server or local storage. This setup enables remote visual monitoring of the farm premises, helping detect unauthorized entry, animal intrusion, or equipment tampering, issues that are often overlooked in traditional farm management systems. The module's compact size, low power consumption, and built-in motion detection capability make it highly suitable for deployment in resource-constrained rural environments. Furthermore, image data from the ESP32-CAM can be paired with automated alerts or integrated with additional ML-based detection models in future upgrades to identify anomalies, enhancing the overall security and situational awareness of the smart farm [54]. This dual use of the ESP32-CAM for both agronomic diagnostics and physical security reinforces the versatility and scalability of the proposed IoT-based precision agriculture platform.

3.5. Limitation and Comparison. Looking ahead, the proposed IoT-ML-PCB-integrated framework holds strong potential for broader deployment in diverse agricultural contexts. Although tomato plants were selected as the case study in this work, the underlying methodology is generalizable and can be extended to a wide variety of crops by retraining or fine-tuning the model with crop-specific datasets. Moreover, the modular hardware and cloud-enabled architecture provide avenues for scaling the system to larger farms, where multiple sensor nodes and image capture units can operate in a coordinated manner. Future research will focus on exploring scalability in extensive

farmlands, enhancing robustness through adaptive learning strategies and incorporating edge-cloud hybrid architectures for real-time analytics. Such advancements will strengthen the applicability of this framework for precision agriculture, resource optimization, and sustainable crop production in both smallholder and industrial-scale farming environments. To contextualize the contribution of our proposed framework, we conducted a comparative review of recent smart agriculture studies that integrate IoT, ML/DL techniques, and automation. Table 2 summarizes the key characteristics of these works, including the ML/DL models used, target crops, notable system features, reported accuracies, and corresponding references. This comparison highlights the progression of research in intelligent farming and positions our approach relative to state-of-the-art solutions. The present system, while validated in controlled experimental conditions, also offers avenues for expansion in more diverse agricultural contexts. For instance, integrating higher resolution sensor arrays and expanding the disease classification dataset to cover a broader range of crops would allow the framework to scale seamlessly across different farming environments. Likewise, adoption of advanced communication protocols such as LoRaWAN or 5G can extend monitoring coverage to large fields, ensuring uninterrupted connectivity even in rural areas. Incorporating renewable energy sources for powering sensor nodes and leveraging edge computing for on-site data processing can further enhance long-term deployability. These directions not only highlight the adaptability of the proposed model but also underscore its capacity to evolve into a comprehensive smart agriculture platform capable of meeting the demands of future precision farming.

4. Conclusion

The primary objective of this study was to monitor the effects of key environmental factors such as temperature, soil moisture, humidity, and atmospheric conditions on plant health and agricultural land. By integrating intelligent sensing, ML, and IoT-based monitoring, the system aims to minimize crop loss and promote the harvesting of healthier crops. Importantly, the study employs noninvasive, safe, and low-risk technologies that pose no threat to public health or environmental safety. Unlike traditional farming practices that heavily rely on manual labor, the proposed system emphasizes a technology-driven approach to agriculture. While such a transition may initially challenge traditional farming communities, the growing global demand for agricultural products necessitates modernization. Technology-enhanced farming offers opportunities for improved yield, greater efficiency, and sustainable resource management, ultimately benefiting farmers in the long term. In contrast to many existing studies that primarily focus on either hardware deployment or software development individually, this work presents a hybrid integration of both hardware and software components through custom PCB designs, IoT communication platforms, and ML algorithms. This unified approach ensures a more robust, scalable, and intelligent agricultural monitoring framework. Although the proposed

ML model has been trained and validated using tomato plants as a case study, its underlying methodology is not crop-specific. The model is designed to process sensor-derived features that are common across a wide range of crops. With appropriate retraining or fine-tuning on crop-specific datasets, the same framework can be readily extended to other agricultural contexts. Therefore, this study not only demonstrates the effectiveness of IoT-ML integration for tomato cultivation but also provides a promising avenue for future research on diverse crops. Although many farmers are currently unfamiliar with advanced agricultural tools, this study emphasizes user-friendly system design to encourage acceptance and adoption. By fostering greater interest in smart farming practices, the proposed approach can contribute to higher annual yields, improved farming efficiency, and enhanced food security, providing substantial benefits to the agriculture-dependent population.

Data Availability Statement

The datasets used in this study are publicly accessible: Cassava Leaf Disease Classification dataset: <https://www.kaggle.com/competitions/cassava-leaf-disease-classification/data>. PlantVillage dataset: <https://www.kaggle.com/datasets/emmarex/plantdisease>. Detailed codes of the ML pipeline and IoT firmware are given in the supporting information.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Md Jahidul Hoq Emon, Sheikh Munim Hussain, Azazul Islam, and Shafin Shadman Ahmed contributed equally to this work.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*)

Figure S1 shows the PCB layout and 3D rendering of Board 1, illustrating the physical arrangement of components and connections used in the environmental monitoring setup.

Figure S2 displays the PCB layout and 3D rendering of Board 2, providing visual insights into its structural and electrical design for optimized sensor integration.

Figure S3: Photograph of the fabricated and packaged Board 1, including connected sensors and power supply, demonstrating the real-world implementation of the design.

Figure S4: Photograph of the fabricated and packaged Board 2, showing the complete setup with sensors and power source, ready for field deployment.

Table S1 presents a comprehensive cost breakdown of all components used in the system, including quantity, unit price in BDT and USD, and total cost. This helps evaluate the economic feasibility of the proposed design.

4. Code: Detailed codes of the ML pipeline and IoT firmware.

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